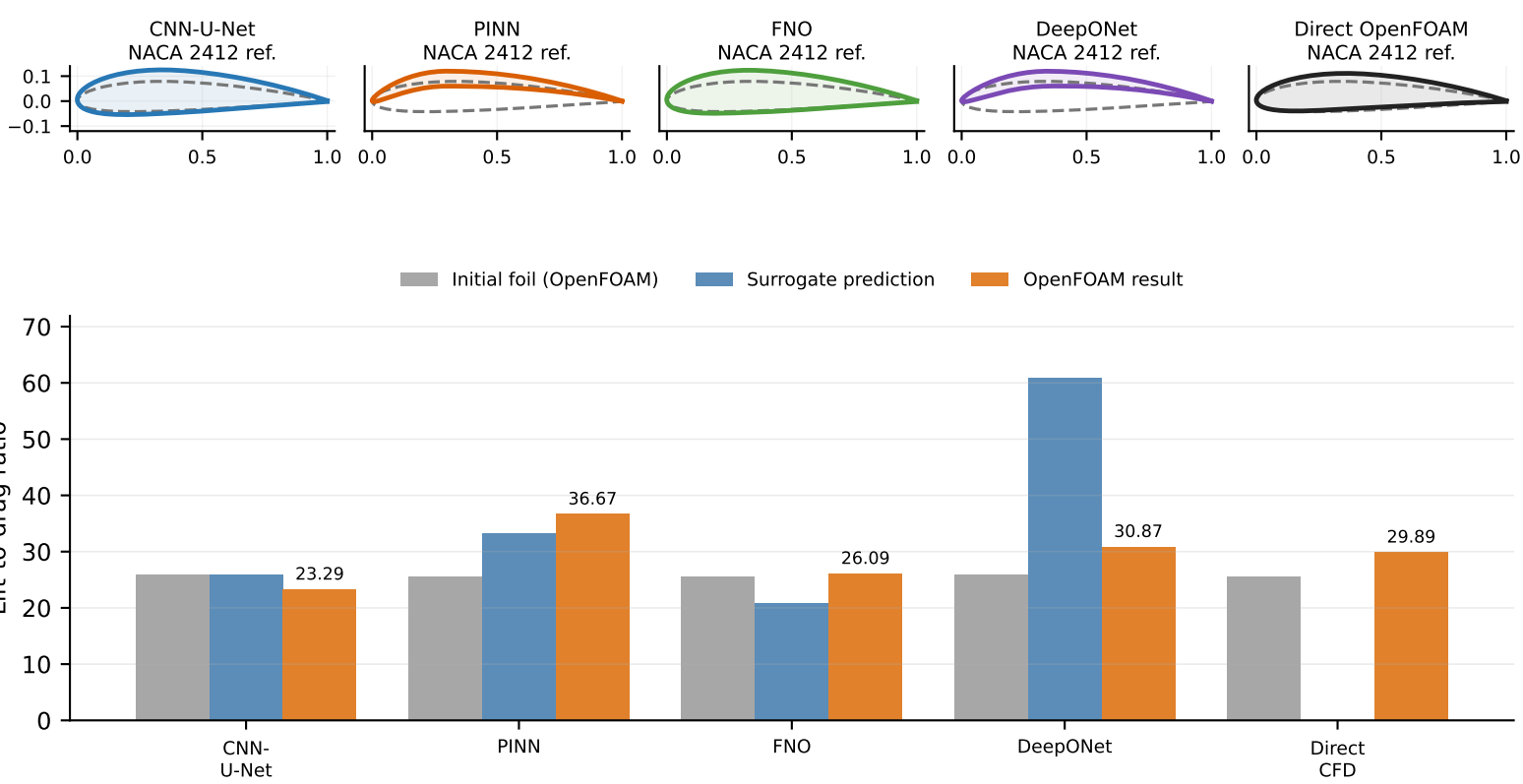
CNN-U-Net
NACA 2412 ref.
PINN
NACA 2412 ref.
FNO
NACA 2412 ref.
DeepONet
NACA 2412 ref.
Direct OpenFOAM
NACA 2412 ref.
0.1
0.0
−0.1
0.0
0.5
1.0
Initial foil (OpenFOAM)
Surrogate prediction
OpenFOAM result
Lift-to-drag ratio
70
60
50
40
30
20
10
0
23.29
36.67
26.09
30.87
29.89
CNN-
U-Net
PINN
FNO
DeepONet
Direct
CFD